Product Requirements Document: Embedded Weather Station and User Interface System

By Billgate Odey | Version 2 (updated 10 November 2026)

1. The why of the product

This product tracks the weather and atmospheric pressure and shows the data on a screen. A small microcontroller reads a barometric pressure and temperature sensor, and the user moves through the readings with a joystick. Pressing the joystick opens a page, and pressing it again returns to the menu.

Secondary features, such as flashing displays and lights from libraries, are not part of the current build and are listed under future work.

2. Product users

Users are any people who want to track weather data and atmospheric pressure, such as hobbyists, students and outdoor enthusiasts. Aviation users, such as airlines, remain a long-term target market. The current prototype is not certified for aviation use.

3. Goals and success metrics

Goal. At a minimum, display the current temperature and atmospheric pressure on a screen with an interactive menu navigated by a joystick. The user can switch between pages showing pressure in hectopascals, pressure in millibars in a large font, and temperature in degrees Celsius.

Additional goals now included. Altitude reading, using a sea-level pressure reference that the user sets with the joystick, and a flicker-free display.

Success metrics (targets for testing):

- The menu appears within 3 seconds of power-up.
- Each joystick action is reflected on screen within 300 ms.
- Sensor values refresh every 500 ms, and page changes show no visible flicker.
- Pressure is within 1 hPa and temperature within 1 °C of a trusted reference, which matches the BMP280 datasheet typical accuracy.
- With a correct sea-level reference, the elevation is within about 15 m of the known elevation. One hPa of pressure error is roughly 8 m of elevation error.
- All five pages can be opened and closed using only the joystick.
- The project is delivered with source code on GitHub, a schematic, a PCB layout, a bill of materials and a working prototype.

4. Scope

In scope

- Reading temperature and barometric pressure from a BMP280 sensor over I2C.

- Showing data on a 2.0 inch ST7789 TFT display over SPI.
- A joystick-driven menu using a KY023 analog joystick with push button.
- Pages for pressure (hPa), pressure (mbar, large font), temperature (°C), elevation (m) and sea-level reference.
- Modular firmware written in PlatformIO and managed in Git on GitHub.
- A schematic, PCB layout and breadboard or veroboard prototype.

Out of scope for this semester

- Retrieving sea-level pressure automatically from Wi-Fi or satellite sources.
- Cloud data storage and an advanced graphical interface.
- Weatherproof enclosure design and aviation certification.

Future work

- Flashing display and LED alerts, for example on a rapid pressure drop.
- Sea-level pressure streamed from a meteorological service over Wi-Fi.
- Cloud logging, pressure trend graphs and battery power.

5. System overview

Hardware

- ESP32-S3-DevKitC-1 microcontroller board.
- BMP280 pressure and temperature sensor on I2C (SDA GPIO8, SCL GPIO9) with 4.7 kΩ pull-ups.
- GMT020027P 2.0 inch ST7789 TFT on SPI (CS GPIO4, DC GPIO5, RST GPIO6, MOSI GPIO11, SCLK GPIO12).
- KY023 joystick: VRX on GPIO1, VRY on GPIO2 and push button on GPIO7, powered from 3.3 V.
- Decoupling capacitors on the 3.3 V rail and at each module.

Firmware modules (headers in `include/`, sources in `src/`)

- `config`: all pin numbers and settings.
- `display`: drawing functions with an off-screen buffer.
- `joystick`: debounced, non-blocking input events.
- `sensor`: BMP280 reads every 500 ms.
- `altitude`: barometric formula.
- `pages` and `ui`: page layouts and the menu state machine.
- `main`: start-up and the main loop.

6. User interface and menu flow

The main menu lists five items: Pressure (hPa), Pressure (mbar), Temperature (C), Elevation (m) and Sea Level Ref. Joystick up and down move the highlight, and a press opens the highlighted page. A press on any page returns to the menu. On the Sea Level Ref page, up and down change the reference by 0.1 hPa and left and right change it by 1.0 hPa.

7. Functional requirements

- **FR-1** The device must read temperature data and show it in °C.
- **FR-2** The device must read barometric pressure data and show it in hPa.
- **FR-3** The device must show pressure in mbar in a large font on its own page.
- **FR-4** The user must navigate the menu with the joystick: up and down to move, press to open, press to return.
- **FR-5** Live values must refresh at least every 500 ms while a page is open.
- **FR-6** The user must be able to set the sea-level pressure reference with the joystick, between 900 and 1100 hPa, with a default of 1013.25 hPa.
- **FR-7** The device must calculate and display elevation above sea level from the measured pressure and the sea-level reference.
- **FR-8** The device must show an error if the sensor is not found, and show -- for invalid readings.
- **FR-9** The device must print debug messages over serial at 115200 baud.
- **FR-10** Navigation must be non-blocking, with no long delays in the main loop.
- **FR-11** Page changes must not flicker.
- **FR-12 (future)** The device may stream the sea-level pressure from a meteorological source over Wi-Fi.

8. Non-functional requirements

- **Temperature tolerance.** The device should cope with temperature swings of 20 °C or more in either direction. The BMP280 is rated from -40 to +85 °C, and the display module range must be confirmed from its datasheet.
- **Power.** The prototype runs from 5 V USB with a target below 500 mA. A 12 V power pack remains an option through a 12 V to 5 V converter, but it is not part of this build.
- **Scalability.** New pages or sensors can be added by adding a module and a menu entry without rewriting existing code.
- **Fits many products.** The design uses standard I2C and SPI interfaces and a single documented pin map.
- **Maintainability.** Code uses one consistent style, with a header and source file per part and all settings in one file.
- **Reliability.** The button is debounced and invalid sensor readings are rejected.
- **Version control.** Frequent, meaningful commits on GitHub, with the supervisors given access.
- **Documentation.** Schematic, PCB layout, bill of materials and a README with the pin map.

9. Assumptions, risks and constraints

Assumptions

- The prototype is powered from 5 V USB through the DevKit. A 12 V pack was the original assumption and is now optional, now works on usb.
- The user can obtain a local sea-level pressure value, for example from a weather forecast, to set the reference.
- Testing happens in a normal indoor or mild outdoor environment.

Risks

- **Elevation error from a wrong reference.** One hPa of error is about 8 m. Mitigation: let the user adjust the reference easily and show it on screen.
- **Joystick damaging the ESP32-S3 inputs.** At 5 V its analog outputs could exceed 3.3 V. Mitigation: power it from 3.3 V.
- **Module pinouts differing between suppliers.** Mitigation: check each module against the schematic before building the board.
- **PCB footprint mismatch with the DevKit.** Mitigation: measure the real board before ordering.
- **Running out of time in one semester.** Mitigation: build in stages, one module per commit.
- **Sensor self-heating or supply noise.** Mitigation: decoupling capacitors and sensible placement.

Constraints

- Power delivery.
- Size.
- Time: about a month.
- Atmospheric conditions affecting wireless transmission, which applies only if Wi-Fi is added later.
- Optimal circuit design.

10. Deliverables and milestones

1. Project skeleton and repository.
2. TFT display working.
3. Joystick integrated.
4. BMP280 sensor integrated.
5. Altitude calculation.
6. Pages and menu.
7. Schematic and PCB layout.
8. Breadboard prototype.
9. Final presentation of the working prototype and code, with the GitHub version control.

11. Open questions

- Should Wi-Fi sea-level retrieval be included if time allows?
- What accuracy is required for elevation?
- What operating temperature range must the display module support?

Changes from version 1

- Replaced the sea-level data stream, and the Wi-Fi and satellite requirement, with manual input by joystick, and moved the automatic version to future work.
- Replaced the unclear 13 V power limit with a 5 V USB supply and a 500 mA target.
- Added measurable success metrics, numbered requirements, a system overview, the menu flow, filled-in risks and milestones.
- Moved altitude reading from a possible feature to a requirement.